COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S2
DATE PERFORMED:	10/02/2026	**DATE SUBMITTED:**	10/02/2026
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ACTIVITY NO. **Graphs**

1. PROCEDURE OUTPUT

ILO A

```
Graph adjacency list
(start_vertex, end_vertex, weight):
(0, 2, 4) (0, 1, 2)
(1, 4, 3)
(2, 3, 2)
(3, 1, 4)
(4, 3, 3)
```

Figure1. Output of Adjacency List

ILO B1

```
1:      {2: 0}, {5: 0},
2:      {1: 0}, {5: 0}, {4: 0},
3:      {4: 0}, {7: 0},
4:      {2: 0}, {3: 0}, {5: 0}, {6: 0}, {8: 0},
5:      {1: 0}, {2: 0}, {4: 0}, {8: 0},
6:      {4: 0}, {7: 0}, {8: 0},
7:      {3: 0}, {6: 0},
8:      {4: 0}, {5: 0}, {6: 0},

DFS Order of vertices:
1
5
8
6
7
3
4
2
```

Figure2. Output of BFS

ILO B2

```
1:      {2: 2}, {5: 3},
2:      {1: 2}, {5: 5}, {4: 1},
3:      {4: 2}, {7: 3},
4:      {2: 1}, {3: 2}, {5: 2}, {6: 4}, {8: 5},
5:      {1: 3}, {2: 5}, {4: 2}, {8: 3},
6:      {4: 4}, {7: 4}, {8: 1},
7:      {3: 3}, {6: 4},
8:      {4: 5}, {5: 3}, {6: 1},

BFS Order of vertices:
1
2
5
4
8
3
6
7
```

Figure3. Output of DFS

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. A person wants to visit different locations indicated on a map. He starts from one location (vertex) and wants to visit every vertex until it finishes from one vertex, backtracks, and then explore other vertex from the same vertex. Discuss which algorithm would be most helpful to accomplish this task.

Depth-first search allows backtracking, and searches from one vertex to another. It goes to the farthest distance, explores that node, and repeats the process backwards until it checks all the remaining nodes.

2. Identify the equivalent of DFS in traversal strategies for trees. To efficiently answer this question, provide a graphical comparison, examine pseudocode and code implementation.

DFS is also similar to pre-order traversal. DFS does its searching by visiting adjacent nodes one by one. This is similar to how pre-order traversal works. Pre-order traversal visits the deepest, most left of the tree and traverses from left to right one by one. The only difference is that pre-order traversal only works on trees, while DFS works on both trees and graphs since it can accommodate the presence of a cycle.

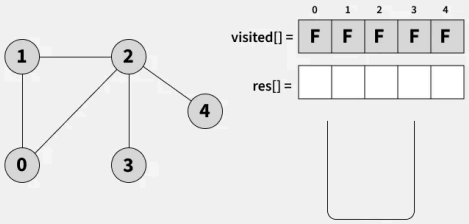
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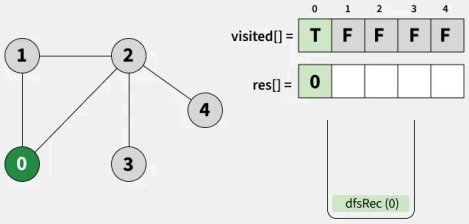
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Depth-First Search

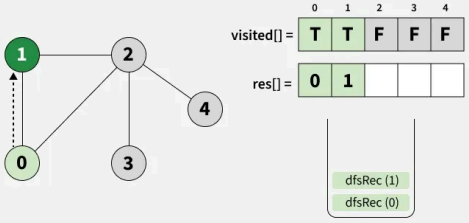
01 Step Maintain a visited array that tracks which nodes have been explored.



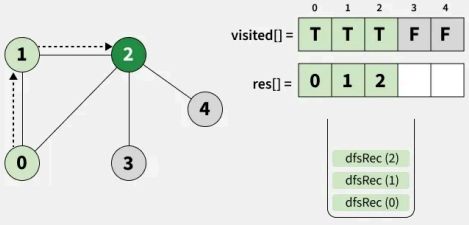
02 Step We start with source vertex 0.



03 Step We visit the neighbour of 0, i.e., 1.

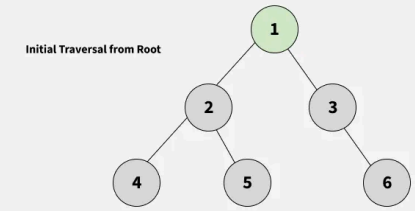


04 Step We visit the neighbour of 1, i.e., 2.

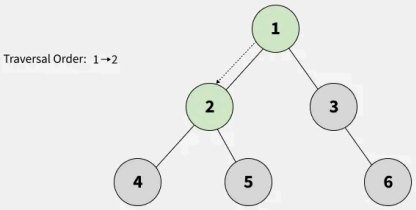


Pre-Order Traversal

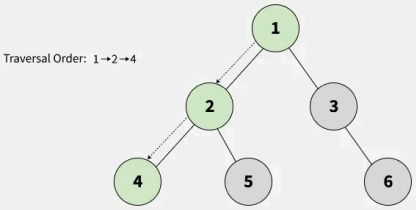
01 Step At first the root will be visited, i.e. node 1.



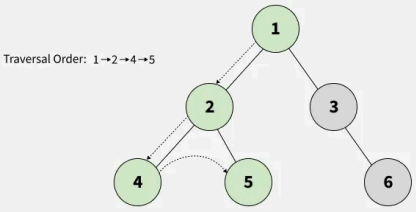
02 Step After this, traverse in the left subtree. Now the root of the left subtree is visited i.e., node 2 is visited.

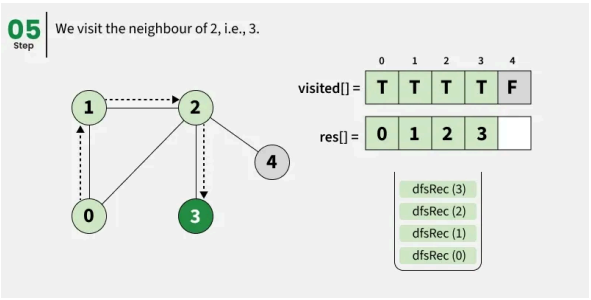


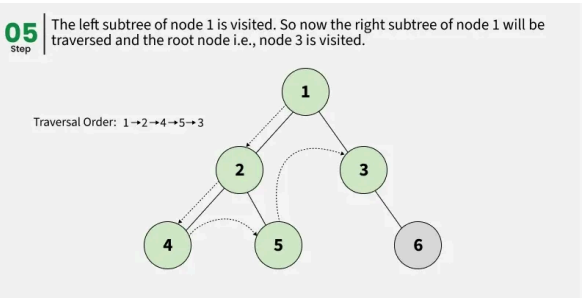
03 Step Again the left subtree of node 2 is traversed and the root of that subtree i.e., node 4 is visited.

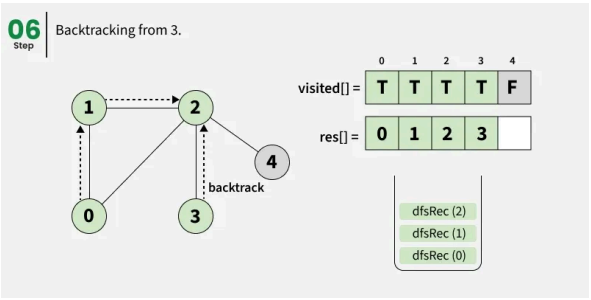


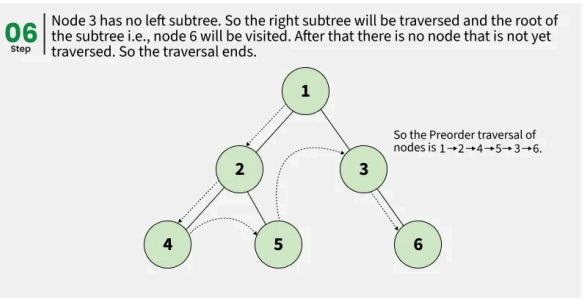
04 Step There is no subtree of 4 and the left subtree of node 2 is visited. So now the right subtree of node 2 will be traversed and the root of that subtree i.e., node 5 will be visited.

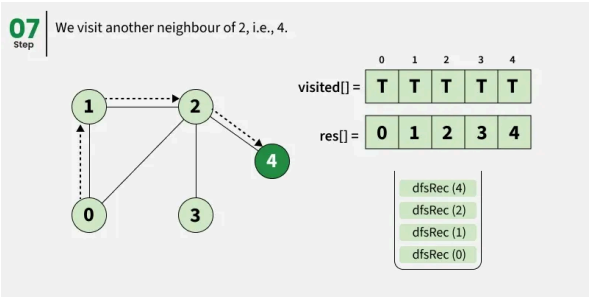


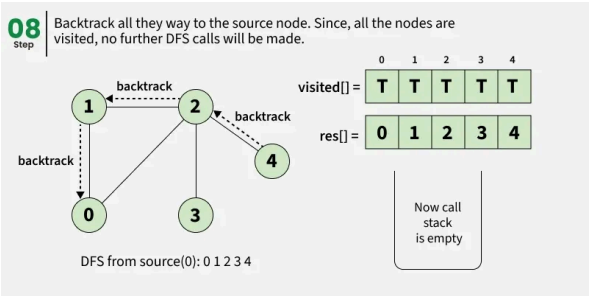












3. In the performed code, what data structure is used to implement the Breadth First Search?

The data structure used to perform or implement the BFS is the Queue data structure. Queue is used because BFS starts from the level of the starting position. Then it checks all the adjacent nodes of the starting node and puts them in a queue list. Then after checking, it checks the next item in the queue and repeats this process until all the items have been checked.

4. How many times can a node be visited in the BFS?

A node can only be visited once in a BFS since it is put into a dequeued state once it is already visited. Making it impossible to be visited again unless it is re-enqueued.

Conclusion:

DFS is a better option when doing a depth based search since it tracks every node from left to right from the bottom. The opposite is for BFS. It starts from the top to the bottom. They both have the same time complexity however, they differ in usage since their efficiency depends on how they will be applied. Overall, I learned a lot in this activity especially in analyzing which is better in a certain setting and how similar they are to other traversal techniques and what makes them special especially when dealing with graphs.

References:

https://www.geeksforgeeks.org/dsa/introduction-to-backtracking-2/
https://www.geeksforgeeks.org/dsa/depth-first-search-or-dfs-for-a-graph/
https://www.geeksforgeeks.org/dsa/preorder-traversal-of-binary-tree/

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I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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